Module 2: Knowledge Representation and Reasoning (KRR)

4.1 Introduction

Artificial Intelligence (AI) aims to develop machines that can think, learn, understand information, and make intelligent decisions similar to humans. Science fiction movies and novels often portray AI-powered robots capable of performing complex tasks intelligently.

However, creating machines that behave exactly like humans is challenging because human intelligence is developed through knowledge, experiences, memories, learning, and conscience.

To make machines intelligent, it is necessary to provide them with knowledge in a form that they can understand and use effectively. This process is called **Knowledge Representation**.

Knowledge and Intelligence

Knowledge is one of the most important components of intelligence.

An intelligent agent:

1. Senses its environment.
2. Uses knowledge about the environment.
3. Makes decisions.
4. Performs appropriate actions.

Relationship between Knowledge and Intelligence

Environment → Knowledge → Decision Making → Intelligent Action

Without knowledge:

- An agent cannot correctly understand its environment.
- It cannot reason effectively.
- It cannot make intelligent decisions.

Therefore,

Knowledge is the foundation of intelligence.

Example

Consider a self-driving car:

- It senses roads, vehicles, pedestrians, and traffic signals.
- Uses stored knowledge about driving rules.
- Makes decisions such as stopping, turning, or accelerating.
- Performs intelligent driving actions.

Without knowledge, the car would be unable to drive safely.

4.2 Knowledge Representation

Definition

Knowledge Representation (KR) is a branch of Artificial Intelligence that deals with representing knowledge in a form that a computer system can understand and use for reasoning and problem-solving.

Knowledge Representation and Reasoning (KRR) focuses on:

- How AI agents think.
- How knowledge is stored.
- How reasoning is performed.
- How intelligent behavior is produced.

Knowledge-Based System

A **Knowledge-Based System (KBS)** stores information about the real world in a structured form.

This enables computers to:

- Understand information.
- Draw conclusions.
- Solve problems.
- Communicate with humans.

Applications

Knowledge representation is used in:

1. Medical diagnosis systems.
2. Expert systems.

3. Natural language processing.
4. Robotics.
5. Self-driving vehicles.
6. Intelligent assistants.

Purpose of Knowledge Representation

The main objectives are:

1. Store Knowledge

Store information about the world.

2. Support Reasoning

Allow machines to think and derive conclusions.

3. Model Intelligent Behavior

Enable machines to behave intelligently.

4. Learn from Experience

Help machines improve performance using previous knowledge.

Important Point

Knowledge representation is **not merely storing data in a database**.

It goes beyond storage by enabling:

- Learning.
- Reasoning.
- Decision making.
- Problem solving.

Thus, machines can behave intelligently like humans.

4.2.1 What Knowledge Needs to be Represented?

Human intelligence consists of more than facts.

Humans use:

- Intuition

- Beliefs
- Intentions
- Judgments
- Common sense
- Experiences

To create intelligent AI systems, all relevant knowledge must be represented in a machine-understandable form.

The following types of knowledge need to be represented.

1. Object Knowledge

Definition

Information and facts about objects present in the environment.

Example

For a self-driving car:

- Vehicles
- Roads
- Traffic signals
- Pedestrians

Importance

The AI system must know:

- What the object is.
- Its characteristics.
- Its role in the environment.

2. Event Knowledge

Definition

Knowledge about actions or events occurring in the real world.

Example

In a self-driving car:

- A pedestrian crossing the road.
- A vehicle suddenly stopping.
- Applying brakes.

Importance

Helps the system understand changing situations and react appropriately.

3. Performance Knowledge

Definition

Knowledge describing **how actions should be performed**.

Example

A robot may know:

- How to pick an object.
- How to move safely.
- How to assemble components.

Importance

Describes behavior and execution methods.

4. Meta-Knowledge

Definition

Knowledge about knowledge.

It refers to understanding:

- What knowledge exists.
- Where it is stored.
- How it should be used.

Example

An expert system knowing which rule should be applied in a particular situation.

Importance

Improves reasoning efficiency.

5. Facts

Definition

Facts are statements that are true about the real world.

Examples

- Water boils at 100°C.
- Earth revolves around the Sun.
- Traffic light red means stop.

Importance

Facts form the foundation of reasoning.

6. Knowledge Base

Definition

A **Knowledge Base (KB)** is a collection of facts, rules, and information stored in a machine-readable form.

It is the most important component of a knowledge-based system.

Characteristics

- Stores knowledge systematically.
- Contains logical sentences.
- Supports reasoning.

Example

Medical Expert System Knowledge Base:

- Symptoms
- Diseases
- Treatment rules

4.2.2 What is Knowledge?

Definition

Knowledge is the fundamental element of human intelligence.

It helps people:

- Understand situations.
- Solve problems.
- Make decisions.
- Learn from experiences.

Knowledge is acquired through:

- Facts
- Data
- Experiences
- Observations
- Situations

Important Statement

A knowledgeable person performs actions more effectively and intelligently.

Types of Knowledge

Every individual possesses different forms of knowledge.

There are **five major types of knowledge**:

1. Meta Knowledge
2. Heuristic Knowledge
3. Procedural Knowledge
4. Declarative Knowledge
5. Structural Knowledge

1. Meta Knowledge

Definition

Meta Knowledge is **knowledge about knowledge**.

It describes:

- What knowledge is available.
- How knowledge should be used.

- When knowledge should be applied.

Example

An AI system knowing which rule is appropriate for solving a particular problem.

Importance

- Improves decision-making.
- Helps select relevant knowledge.
- Increases reasoning efficiency.

2. Heuristic Knowledge

Definition

Heuristic Knowledge is specialized knowledge gained from:

- Experience
- Expertise
- Practical observations

It is commonly known as a **Rule of Thumb**.

Characteristics

- Experience-based
- Practical
- Not guaranteed to be correct
- Often produces good solutions quickly

Example

A doctor's experience suggesting that certain symptoms may indicate a specific disease.

AI Example

Chess programs use heuristic rules to select promising moves.

Importance

- Reduces search time.
- Helps solve complex problems quickly.

- Supports expert systems.

3. Procedural Knowledge

Definition

Procedural Knowledge describes:

How to perform a task.

It is also called **Imperative Knowledge**.

Contains

- Procedures
- Rules
- Strategies
- Algorithms
- Plans

Examples

- How to drive a car.
- How to cook rice.
- How to install software.

AI Example

A robot's instructions for picking and placing an object.

Importance

Provides step-by-step guidance for performing tasks.

4. Declarative Knowledge

Definition

Declarative Knowledge provides information about:

- Facts
- Objects
- Concepts

- Properties

It answers:

"What is it?"

It is also called **Descriptive Knowledge**.

Characteristics

- Easier to represent.
- Expressed through statements.
- Describes facts and attributes.

Examples

- Delhi is the capital of India.
- A triangle has three sides.
- A car has four wheels.

AI Example

Database facts stored in an expert system.

Importance

Provides descriptive information about the world.

Declarative Knowledge	**Procedural Knowledge**
Describes facts	Describes procedures
Answers "What?"	Answers "How?"
Static information	Action-oriented information
Example: Earth is a planet	Example: How to launch a satellite

5. Structural Knowledge

Definition

Structural Knowledge describes relationships among concepts and objects.

It explains:

- How concepts are connected.
- Organization of knowledge.
- Dependencies between concepts.

Relationship Types

a) Kind-of Relationship

Shows category hierarchy.

Example:

Animal → Dog

Dog is a kind of Animal.

b) Part-of Relationship

Shows component relationships.

Example:

Car → Engine

Engine is part of a Car.

c) Grouping Relationship

Shows membership within a collection.

Example:

Students belong to a classroom.

Importance

Structural knowledge helps in:

- Understanding relationships.
- Organizing knowledge.
- Solving complex problems efficiently.

Types of Knowledge Representation (Easy Explanation)

Knowledge Representation (KR) is a way of storing knowledge in a computer so that it can understand, reason, and make decisions.

1. Simple Relational Knowledge

This method stores facts in the form of **tables (relations)**, similar to a database.

Example

Student ID	Name	Course
101	Diya	BCA
102	Ravi	BSc

Here, each row represents an object (student), and columns represent attributes.

Advantages

- Easy to store and retrieve data.
- Similar to database tables.

Disadvantages

- Limited reasoning capability.
- Difficult to derive new knowledge automatically.

2. Inheritable Knowledge

In this method, knowledge is organized in a **hierarchy of classes and subclasses**.

Example

Student
|
|-- Undergraduate Student
| |
| |-- Zinya
|
|-- Postgraduate Student
|
|-- Vidhi

Explanation

- **Student** is a general class.

- **Undergraduate Student** and **Postgraduate Student** are specialized classes.
- **Zinya** and **Vidhi** are instances (individual objects).

This relationship is called an **IS-A relationship**.

- Zinya **IS-A** Undergraduate Student.
- Undergraduate Student **IS-A** Student.

Advantages

- Reduces repetition of information.
- Properties can be inherited from parent classes.

Example

If all students have an ID card, then:

- Undergraduate students automatically have an ID card.
- Postgraduate students automatically have an ID card.

3. Inferential Knowledge

This method represents knowledge using **logic** and allows the system to derive new facts.

Given Statements

Statement 1: Diya is a student.

Student(Diya)

Statement 2: All students are bright.

∀x [Student(x) → Bright(x)]

Inference

Since:

- Diya is a student.
- All students are bright.

Therefore,

Bright(Diya)

Explanation

The system did not store "Diya is bright" directly. It **inferred** (derived) this fact using logical reasoning.

Advantages

- Supports intelligent decision-making.
- Can generate new knowledge from existing facts.

Disadvantages

- More complex to implement.
- Requires logical rules.

4.5 Techniques of Knowledge Representation in AI – Notes

Introduction

Knowledge Representation (KR) is the method of storing knowledge in a form that a computer can understand and use for reasoning, problem-solving, and decision-making.

Techniques of Knowledge Representation

1. Logical Representation
2. Semantic Network Representation
3. Frame Representation
4. Production Rules
5. Propositional Logic

4.5.1 Logical Representation

Definition

Logical Representation uses a formal language with predefined rules to represent knowledge without ambiguity. It helps in drawing conclusions through logical reasoning.

Components

Syntax

- Defines the rules for constructing valid logical statements.

Semantics

- Defines the meaning and interpretation of logical statements.

Types of Logical Representation

1. Propositional Logic

- Uses propositions that are either True or False.
- Example:
    - P: It is raining.
    - Q: The road is wet.

2. First Order Logic (FOL)

- More expressive than propositional logic.
- Uses objects, predicates, and quantifiers.
- Example:
    - Student(Diya)
    - ∀x [Student(x) → Bright(x)]

Advantages

- Precise and unambiguous.
- Supports logical reasoning.
- Easy to verify conclusions.

Disadvantages

- Difficult to represent complex real-world knowledge.
- Computationally expensive.
- Less natural for humans.

4.5.2 Semantic Network Representation

Definition

A Semantic Network represents knowledge using a graphical structure consisting of nodes and links.

Components

Nodes

Represent objects, concepts, or entities.

Arcs (Links)

Represent relationships between objects.

Types of Relations

IS-A Relation

Represents inheritance.

Example:

- Parrot IS-A Bird
- Bird IS-A Animal

Instance Relation

Represents membership.

Example:

- Mittu is an instance of Parrot.

Example

Animal
|
IS-A
|
Bird
|
IS-A
|
Parrot

Advantages

- Easy to understand and visualize.

- Supports inheritance.
- Natural representation of knowledge.

Disadvantages

- Difficult to manage large networks.
- Limited reasoning capabilities.
- Relationships may become complex.

4.5.3 Frame Representation

Definition

A Frame is a record-like structure used to represent knowledge about an object or event through attributes and values.

Components

Slot

An attribute of an object.

Facet

The value associated with a slot.

Example: Frame for a Book

Slot	Value
Title	AI Fundamentals
Author	John Smith
Publisher	ABC Publications
Price	₹500
Pages	350

Characteristics

- Derived from semantic networks.
- Similar to classes and objects in object-oriented programming.

- Frames can be interconnected.

Advantages

- Structured representation.
- Easy inheritance of properties.
- Efficient storage of object information.

Disadvantages

- Difficult to handle complex reasoning.
- Maintenance becomes difficult for large systems.

4.5.4 Production Rules

Definition

Production Rules represent knowledge in the form of IF–THEN statements.

General Form

IF Condition
THEN Action

Examples

Example 1

IF Hungry
THEN Eat

Example 2

IF On Bus AND Bought Ticket AND Vacant Seat
THEN Sit Down

Components of a Production Rule System

1. Set of Production Rules

Collection of IF–THEN rules.

2. Working Memory

Stores the current state of the problem.

3. Recognize–Act Cycle

- Recognize applicable rules.
- Select a rule.
- Execute the corresponding action.

Conflict Resolution

When multiple rules are applicable simultaneously, the system selects the most appropriate rule. This situation is called a Conflict Set.

Advantages

- Simple and easy to understand.
- Modular structure.
- Widely used in expert systems.

Disadvantages

- Large rule sets may be required.
- Conflict resolution can become complex.

4.5.5 Propositional Logic (PL)

Definition

Propositional Logic is the simplest form of logic in which knowledge is represented using propositions.

Proposition

A declarative statement that is either True or False, but not both.

Examples

True Proposition

2 + 2 = 4

False Proposition

The Sun rises in the west.

Important Concepts

Atomic Proposition

A simple proposition consisting of a single statement.

Examples:

- 2 + 2 = 4
- The Sun rises in the east

Compound Proposition

Formed by combining atomic propositions using logical operators.

Examples:

- The Sun rises in the east AND sets in the west.
- It is raining OR it is cloudy.

Logical Operators

Symbol	Meaning
¬P	NOT P
P ∧ Q	P AND Q
P ∨ Q	P OR Q
P → Q	IF P THEN Q
P ↔ Q	IF AND ONLY IF

Tautology

A proposition that is always true.

Example:

P ∨ ¬P

Contradiction

A proposition that is always false.

Example:

P ∧ ¬P

Satisfiable Proposition

A proposition that is true for at least one interpretation.

Example:

P ∨ Q

Contingent Proposition

A proposition that is neither always true nor always false.

Example:

I am not well, so I am taking leave today.

Statements That Are Not Propositions

Questions

- Which course are you opting for?

Commands

- Read the newspaper.

Opinions

- I think it is fiction.

These statements cannot be assigned definite True or False values.

4.6 Syntax of Propositional Logic

Definition

Syntax specifies the rules for constructing valid propositional logic expressions.

Types of Propositions

1. Atomic Proposition

A single indivisible statement.

Examples:

- 2 + 2 = 4
- The Sun rises in the west

2. Compound Proposition

A combination of atomic propositions connected using logical operators.

Examples:

- The Sun rises in the east AND sets in the west.
- Ruskin Bond authored children's books AND received the Sahitya Akademi Award.